Combined Bioassays 2025

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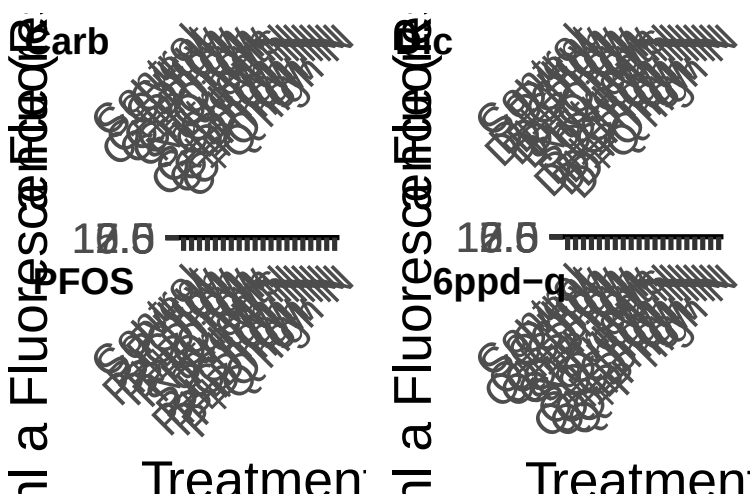
2026-02-11

Boxplots for each bioassay

```
############# figures:

Carb_box = ggplot() +
  geom_boxplot(aes(x = Group,
                   y = `72h`),
               fill = "lightblue4",
               data = Carb_RFU) +
  xlab("Treatment") +
  ylab("Chl a Fluorescence (RFU)") +
  scale_y_continuous(limits = c(1, 12)) +
  theme_classic(base_size = 20) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(Carb_box, filename = "Figures/end_RFU/Carb_box.png",
       device = "png", height = 7, width = 11)

Dic_box = ggplot() +
  geom_boxplot(aes(x = Group,
                   y = `72h`),
               fill = "lightblue4",
               data = Dic_RFU) +
  xlab("Treatment") +
  ylab("Chl a Fluorescence (RFU)") +
  scale_y_continuous(limits = c(1, 12)) +
  theme_classic(base_size = 20) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(Dic_box, filename = "Figures/end_RFU/Dic_box.png",
       device = "png", height = 7, width = 11)

PFOS_box = ggplot() +
  geom_boxplot(aes(x = Group,
                   y = `72h`),
               fill = "lightblue4",
               data = PFOS_RFU) +
  xlab("Treatment") +
  ylab("Chl a Fluorescence (RFU)") +
  scale_y_continuous(limits = c(1, 12)) +
  theme_classic(base_size = 20) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(PFOS_box, filename = "Figures/end_RFU/PFOS_box.png",
```

```
       device = "png", height = 7, width = 11)

Q_box = ggplot() +
  geom_boxplot(aes(x = Group,
                   y = `72h`),
               fill = "lightblue4",
               data = Q_RFU) +
  xlab("Treatment") +
  ylab("Chl a Fluorescence (RFU)") +
  scale_y_continuous(limits = c(1, 12)) +
  theme_classic(base_size = 20) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(Q_box, filename = "Figures/end_RFU/Q_box.png",
       device = "png", height = 7, width = 11)



combined = ggarrange(Carb_box, Dic_box, PFOS_box, Q_box,
          ncol = 2, nrow = 2,
          labels = c("Carb", "Dic", "PFOS", "6ppd-q"),
          common.legend = TRUE,
          legend = "bottom")


print(combined)
```



Percent Difference from Control (Solvent)

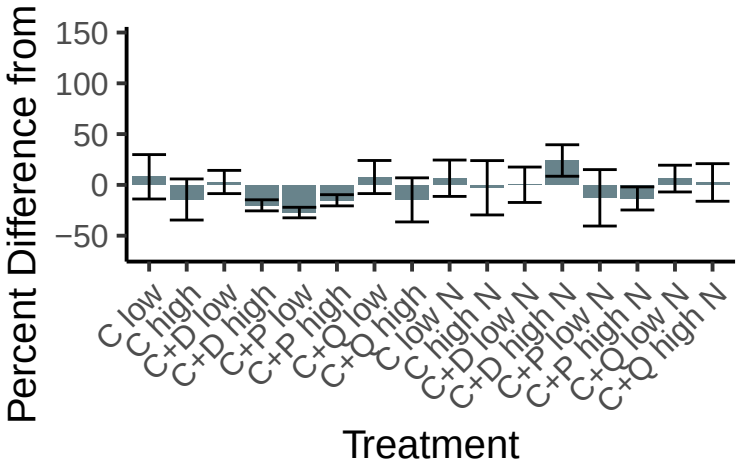
```
############# import data:

pd = read_excel("2025_combinations_master.xlsx", sheet = "all_pd")
glimpse(pd)
```

```
## Rows: 16
## Columns: 12
## $ Carbamazepine <chr> "Carblow", "Carbhigh", "Carb+Diclow", "Carb+Dichigh", "C~
## $ avg_pd_C      <dbl> 7.971586, -14.364641, 2.841358, -20.126283, -27.229676, ~
```

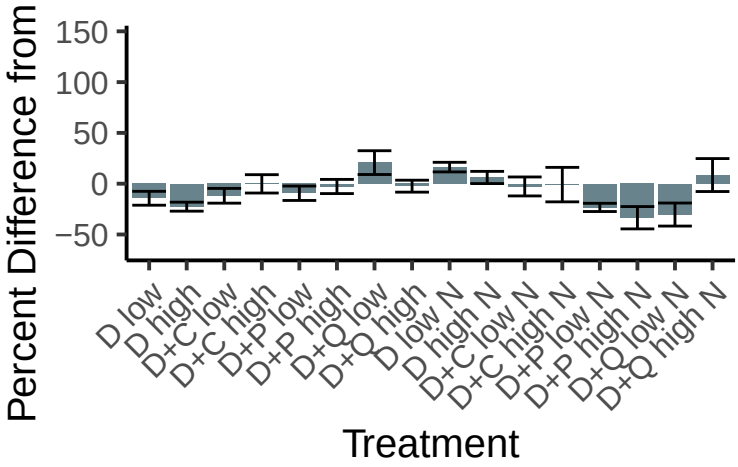
```
## $ sd_pd_C       <dbl> 21.849984, 20.223904, 11.430650, 5.449937, 5.184382, 5.5~
## $ Diclofenac    <chr> "Diclow", "Dichigh", "Dic+Carblow", "Dic+Carbhigh", "Dic~
## $ avg_pd_D      <dbl> -14.3435981, -22.6094003, -11.9124797, -0.1620746, -9.40~
## $ sd_pd_D       <dbl> 6.837293, 4.431685, 7.259662, 9.033627, 7.104217, 6.9924~
## $ PFOS          <chr> "PFOSlow", "PFOShigh", "PFOS+Carblow", "PFOS+Carbhigh", ~
## $ avg_pd_P      <dbl> 11.553785, -6.374502, 11.633466, 6.454183, -11.474104, 3~
## $ sd_pd_P       <dbl> 30.367662, 16.007014, 12.918855, 8.540405, 12.694105, 11~
## $ Quinone       <chr> "6ppdqlow", "6ppdqhigh", "6ppdq+Carblow", "6ppdq+Carbhig~
## $ avg_pd_Q      <dbl> -21.862668, -34.096290, -17.205998, -26.756117, -32.9123~
## $ sd_pd_Q       <dbl> 8.979613, 16.254792, 16.839072, 7.625034, 13.046227, 11.~
```

```
############# rearrange and reformat data:


pd$Carbamazepine <- factor(pd$Carbamazepine,
                              levels = c("Carblow", "Carbhigh", "Carb+Diclow", "Carb+Dichigh", "Carb+PFO
                              labels = c("C low", "C high", "C+D low", "C+D high", "C+P low", "C+P high

pd$Diclofenac <- factor(pd$Diclofenac,
                            levels = c("Diclow", "Dichigh", "Dic+Carblow", "Dic+Carbhigh", "Dic+PFOSlow
                            labels = c("D low", "D high", "D+C low", "D+C high", "D+P low", "D+P high"

pd$PFOS <- factor(pd$PFOS,
                               levels = c("PFOSlow", "PFOShigh", "PFOS+Carblow", "PFOS+Carbhigh", "PFOS+
                               labels = c("P low", "P high", "P+C low", "P+C high", "P+D low", "P+D high

pd$Quinone <- factor(pd$Quinone,
                     levels = c("6ppdqlow", "6ppdqhigh", "6ppdq+Carblow", "6ppdq+Carbhigh", "6ppdq+Dicl
                     labels = c("Q low", "Q high", "Q+C low", "Q+C high", "Q+D low", "Q+D high", "Q+P l


############# figures:

pd_carb = ggplot() +
  geom_col(aes(x = Carbamazepine,
               y = avg_pd_C),
           fill = "lightblue4",
           data = pd) +
  geom_errorbar(data = pd, aes(x = Carbamazepine,
                               y = avg_pd_C,
                               ymin = avg_pd_C-sd_pd_C,
                               ymax = avg_pd_C+sd_pd_C)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-65, 145)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_carb, filename = "Figures/percent_difference_control/pd_carb.png",
       device = "png", height = 7, width = 11)

pd_dic = ggplot() +
  geom_col(aes(x = Diclofenac,
               y = avg_pd_D),
           fill = "lightblue4",
```

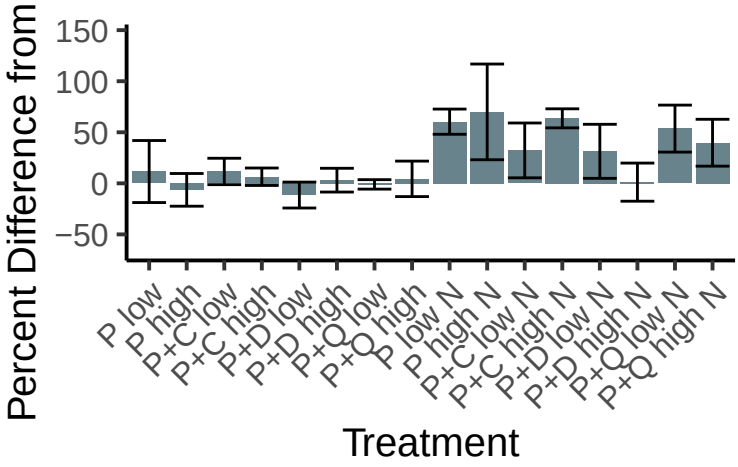
```
           data = pd) +
  geom_errorbar(data = pd, aes(x = Diclofenac,
                               y = avg_pd_D,
                               ymin = avg_pd_D-sd_pd_D,
                               ymax = avg_pd_D+sd_pd_D)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-65, 145)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_dic, filename = "Figures/percent_difference_control/pd_dic.png",
       device = "png", height = 7, width = 11)

pd_PFOS = ggplot() +
  geom_col(aes(x = PFOS,
               y = avg_pd_P),
           fill = "lightblue4",
           data = pd) +
  geom_errorbar(data = pd, aes(x = PFOS,
                                    y = avg_pd_P,
                                    ymin = avg_pd_P-sd_pd_P,
                                    ymax = avg_pd_P+sd_pd_P)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-65, 145)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_PFOS, filename = "Figures/percent_difference_control/pd_PFOS.png",
       device = "png", height = 7, width = 11)

pd_Q = ggplot() +
  geom_col(aes(x = Quinone,
               y = avg_pd_Q),
           fill = "lightblue4",
           data = pd) +
  geom_errorbar(data = pd, aes(x = Quinone,
                               y = avg_pd_Q,
                               ymin = avg_pd_Q-sd_pd_Q,
                               ymax = avg_pd_Q+sd_pd_Q)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-65, 145)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_Q, filename = "Figures/percent_difference_control/pd_Q.png",
       device = "png", height = 7, width = 11)

pd_carb
```



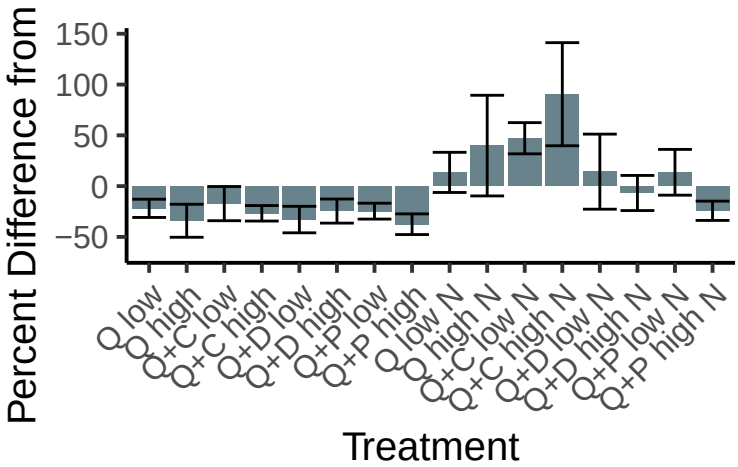

```
pd_dic
```




```
pd_PFOS
```



```r
pd_Q
```




```r
############# import data:

delta = read_excel("2025_combinations_master.xlsx", sheet = "all_delta")

glimpse(delta)
```

Change in RFU over 72h

```
## Rows: 20
## Columns: 12
## $ Carbamazepine <chr> "Control", "Solvent", "nutrients only", "nutrients + sol~
## $ avg_delta_C   <dbl> -0.7950, -1.1900, 4.7625, 4.2075, -0.5025, -1.1325, -0.5~
## $ sd_delta_C    <dbl> 0.4542026, 0.6423395, 0.8682694, 1.0930195, 0.6080228, 0~
## $ Diclofenac    <chr> "Control", "Solvent", "nutrients only", "nutrients + sol~
## $ avg_delta_D   <dbl> -0.5475, -0.4100, 2.9825, 3.4900, -0.7725, -1.0275, -0.7~
## $ sd_delta_D    <dbl> 0.1510794, 0.5460769, 0.8328815, 0.8481745, 0.2182315, 0~
## $ PFOS          <chr> "Control", "Solvent", "nutrients only", "nutrients + sol~
## $ avg_delta_P   <dbl> -0.8475, -0.1975, 2.7350, 0.6025, 0.3400, -0.3125, 0.427~
## $ sd_delta_P    <dbl> 0.2355667, 1.0083113, 0.5700000, 0.9389134, 0.9522955, 0~
## $ Quinone       <chr> "Control", "Solvent", "nutrients only", "nutrients + sol~
## $ avg_delta_Q   <dbl> -0.1050, 0.0525, 4.3775, 1.7350, -0.6500, -0.7975, -0.54~
## $ sd_delta_Q    <dbl> 0.25683977, 0.05852350, 0.68182965, 1.07673890, 0.387814~
```

```r
summary(delta)
```

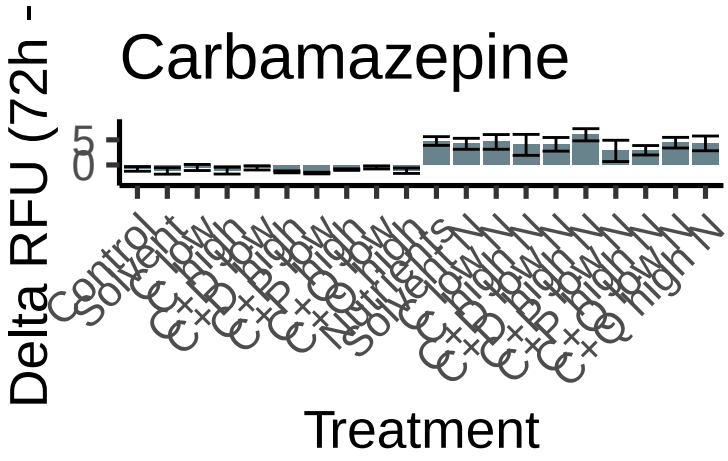
```
##  Carbamazepine       avg_delta_C        sd_delta_C      Diclofenac
##  Length:20          Min.   :-1.6200   Min.   :0.1535   Length:20
##  Class :character   1st Qu.:-0.9788   1st Qu.:0.4448   Class :character
##  Mode  :character   Median : 1.1500   Median :0.7738   Mode  :character
##                     Mean   : 1.6169   Mean   :0.8917
##                     3rd Qu.: 4.2287   3rd Qu.:1.2474
```

```
##                      Max.   : 6.0100   Max.   :2.1229
##   avg_delta_D        sd_delta_D         PFOS            avg_delta_P
##  Min.   :-1.0275   Min.   :0.1320   Length:20          Min.   :-0.8475
##  1st Qu.:-0.4669   1st Qu.:0.3048   Class :character   1st Qu.:-0.0350
##  Median : 0.8575   Median :0.4169   Mode  :character   Median : 0.5150
##  Mean   : 1.2334   Mean   :0.4868                      Mean   : 1.0004
##  3rd Qu.: 3.1606   3rd Qu.:0.6589                      3rd Qu.: 2.1800
##  Max.   : 4.4775   Max.   :1.0110                      Max.   : 3.0600
##    sd_delta_P       Quinone           avg_delta_Q        sd_delta_Q
##  Min.   :0.2223   Length:20          Min.   :-1.0250   Min.   :0.05852
##  1st Qu.:0.3398   Class :character   1st Qu.:-0.6350   1st Qu.:0.38248
##  Median :0.6129   Mode  :character   Median : 0.2888   Median :0.46126
##  Mean   :0.6439                      Mean   : 1.0910   Mean   :0.75039
##  3rd Qu.:0.8902                      3rd Qu.: 2.3056   3rd Qu.:0.90551
##  Max.   :1.8223                      Max.   : 5.4975   Max.   :2.27504
```

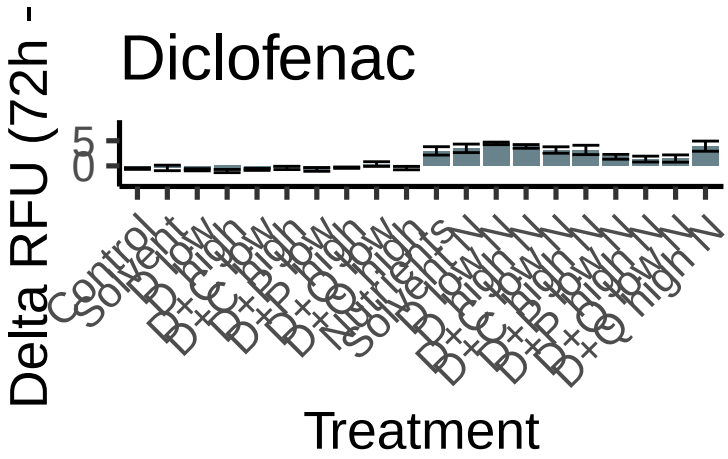
```
############# rearrange and reformat data:


delta$Carbamazepine <- factor(delta$Carbamazepine,
                         levels = c("Control", "Solvent", "Carblow", "Carbhigh", "Carb+Diclow", "Carb+D
                         labels = c("Control", "Solvent", "C low", "C high", "C+D low", "C+D high", "C+

delta$Diclofenac <- factor(delta$Diclofenac,
                        levels = c("Control", "Solvent", "Diclow", "Dichigh", "Dic+Carblow", "Dic+Carbh
                        labels = c("Control", "Solvent", "D low", "D high", "D+C low", "D+C high", "D+P


delta$PFOS <- factor(delta$PFOS,
                         levels = c("Control", "Solvent", "PFOSlow", "PFOShigh", "PFOS+Carblow", "PFOS+
                         labels = c("Control", "Solvent", "P low", "P high", "P+C low", "P+C high", "P+

delta$Quinone <- factor(delta$Quinone,
                      levels = c("Control", "Solvent", "6ppdqlow", "6ppdqhigh", "6ppdq+Carblow", "6ppdq
                      labels = c("Control", "Solvent", "Q low", "Q high", "Q+C low", "Q+C high", "Q+D l

############# Figures:


delta_carb = ggplot() +
  geom_col(aes(x = Carbamazepine,
               y = avg_delta_C),
           fill = "lightblue4",
           data = delta) +
  geom_errorbar(data = delta, aes(x = Carbamazepine,
                                  y = avg_delta_C,
                                  ymin = avg_delta_C-sd_delta_C,
                                  ymax = avg_delta_C+sd_delta_C)) +
  xlab("Treatment") +
  ylab("Delta RFU (72h - T0)") +
  ggtitle("Carbamazepine") +
  theme_classic(base_size = 20) +
  scale_y_continuous(limits = c(-3.5, 8.5)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```r
ggsave(delta_carb, filename = "Figures/delta_RFU/delta_carb.png",
       device = "png", height = 7, width = 11)

delta_dic = ggplot() +
  geom_col(aes(x = Diclofenac,
               y = avg_delta_D),
           fill = "lightblue4",
           data = delta) +
  geom_errorbar(data = delta, aes(x = Diclofenac,
                                  y = avg_delta_D,
                                  ymin = avg_delta_D-sd_delta_D,
                                  ymax = avg_delta_D+sd_delta_D)) +
  xlab("Treatment") +
  ylab("Delta RFU (72h - T0)") +
  ggtitle("Diclofenac") +
  theme_classic(base_size = 20) +
  scale_y_continuous(limits = c(-3.5, 8.5)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(delta_dic, filename = "Figures/delta_RFU/delta_dic.png",
       device = "png", height = 7, width = 11)


delta_PFOS = ggplot() +
  geom_col(aes(x = PFOS,
               y = avg_delta_P),
           fill = "lightblue4",
           data = delta) +
  geom_errorbar(data = delta, aes(x = PFOS,
                                  y = avg_delta_P,
                                  ymin = avg_delta_P-sd_delta_P,
                                  ymax = avg_delta_P+sd_delta_P)) +
  xlab("Treatment") +
  ylab("Delta RFU (72h - T0)") +
  ggtitle("PFOS") +
  theme_classic(base_size = 20) +
  scale_y_continuous(limits = c(-3.5, 8.5)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(delta_PFOS, filename = "Figures/delta_RFU/delta_PFOS.png",
       device = "png", height = 7, width = 11)


delta_Q = ggplot() +
  geom_col(aes(x = Quinone,
               y = avg_delta_Q),
           fill = "lightblue4",
           data = delta) +
  geom_errorbar(data = delta, aes(x = Quinone,
                                  y = avg_delta_Q,
                                  ymin = avg_delta_Q-sd_delta_Q,
                                  ymax = avg_delta_Q+sd_delta_Q)) +
  xlab("Treatment") +
  ylab("Delta RFU (72h - T0)") +
  ggtitle("6ppd-quinone") +
```

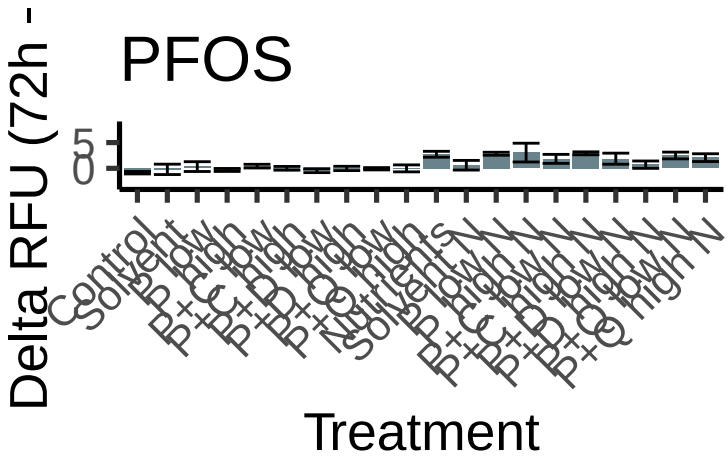
```
  theme_classic(base_size = 20) +
  scale_y_continuous(limits = c(-3.5, 8.5)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(delta_Q, filename = "Figures/delta_RFU/delta_Q.png",
       device = "png", height = 7, width = 11)


combined_delta = ggarrange(delta_carb + rremove("ylab") + rremove("xlab"), delta_dic + rremove("ylab") +
                           labels = c("A", "Diclofenac", "PFOS", "6ppd-quinone"),
                           label.x = 0.05,
                           label.y = 1,
                           nrow = 2,
                           ncol = 2)

combined_delta = ggarrange(delta_carb + rremove("xlab"), delta_dic + rremove("ylab") + rremove("xlab"),
                          #labels = c("A", "Diclofenac", "PFOS", "6ppd-quinone"),
                          label.x = 0.05,
                          label.y = 1,
                          nrow = 2,
                          ncol = 2)



combined_delta  = annotate_figure(combined_delta, left = textGrob(label = expression(Delta * "Chl a Flu
                                  bottom = textGrob("Treatment", gp = gpar(cex = 1.3, fontsize = 18))


delta_carb
```



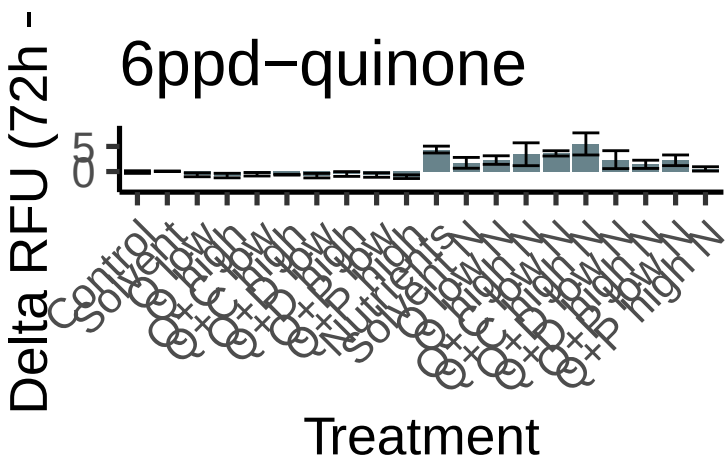

```
delta_dic
```



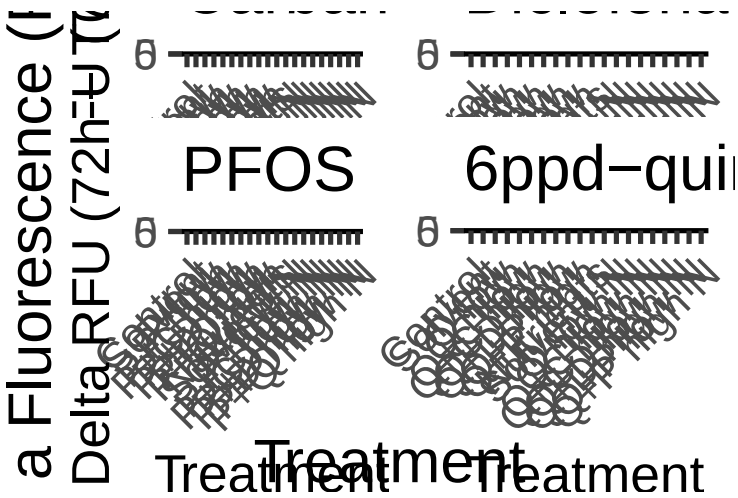

```
delta_PFOS
```




```
delta_Q
```



```
combined_delta
```




```
############# import data:

pd_delta = read_excel("2025_combinations_master.xlsx", sheet = "all_pd_delta")
glimpse(pd)
```

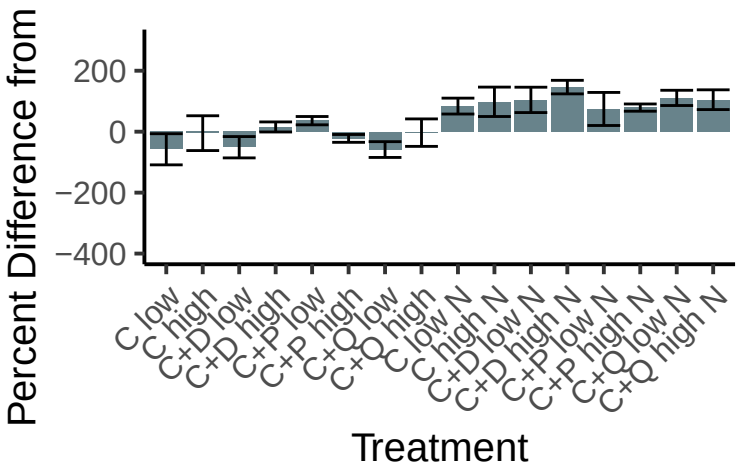
Percent Difference in the Change in RFU over 72h

```
## Rows: 16
## Columns: 12
## $ Carbamazepine <fct> C low, C high, C+D low, C+D high, C+P low, C+P high, C+Q~
## $ avg_pd_C      <dbl> 7.971586, -14.364641, 2.841358, -20.126283, -27.229676, ~
## $ sd_pd_C       <dbl> 21.849984, 20.223904, 11.430650, 5.449937, 5.184382, 5.5~
## $ Diclofenac    <fct> D low, D high, D+C low, D+C high, D+P low, D+P high, D+Q~
## $ avg_pd_D      <dbl> -14.3435981, -22.6094003, -11.9124797, -0.1620746, -9.40~
## $ sd_pd_D       <dbl> 6.837293, 4.431685, 7.259662, 9.033627, 7.104217, 6.9924~
## $ PFOS          <fct> P low, P high, P+C low, P+C high, P+D low, P+D high, P+Q~
## $ avg_pd_P      <dbl> 11.553785, -6.374502, 11.633466, 6.454183, -11.474104, 3~
## $ sd_pd_P       <dbl> 30.367662, 16.007014, 12.918855, 8.540405, 12.694105, 11~
## $ Quinone       <fct> Q low, Q high, Q+C low, Q+C high, Q+D low, Q+D high, Q+P~
## $ avg_pd_Q      <dbl> -21.862668, -34.096290, -17.205998, -26.756117, -32.9123~
## $ sd_pd_Q       <dbl> 8.979613, 16.254792, 16.839072, 7.625034, 13.046227, 11.~
```

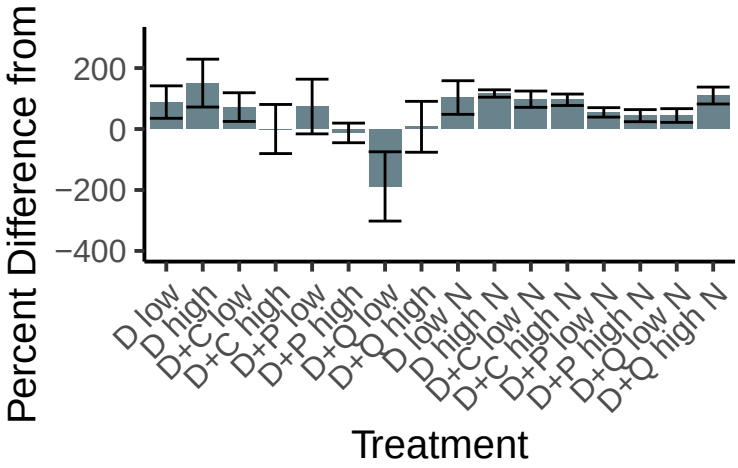
```
############# rearrange and reformat data:


pd_delta$Carbamazepine <- factor(pd_delta$Carbamazepine,
                            levels = c("Carblow", "Carbhigh", "Carb+Diclow", "Carb+Dichigh", "Carb+PFOSl
                            labels = c("C low", "C high", "C+D low", "C+D high", "C+P low", "C+P high",

pd_delta$Diclofenac <- factor(pd_delta$Diclofenac,
                          levels = c("Diclow", "Dichigh", "Dic+Carblow", "Dic+Carbhigh", "Dic+PFOSlow", "
                          labels = c("D low", "D high", "D+C low", "D+C high", "D+P low", "D+P high", "D+
```

```r
pd_delta$PFOS <- factor(pd_delta$PFOS,
                   levels = c("PFOSlow", "PFOShigh", "PFOS+Carblow", "PFOS+Carbhigh", "PFOS+Diclow", "PFO
                   labels = c("P low", "P high", "P+C low", "P+C high", "P+D low", "P+D high", "P+Q low"

pd_delta$Quinone <- factor(pd_delta$Quinone,
                      levels = c("6ppdqlow", "6ppdqhigh", "6ppdq+Carblow", "6ppdq+Carbhigh", "6ppdq+Dicl
                      labels = c("Q low", "Q high", "Q+C low", "Q+C high", "Q+D low", "Q+D high", "Q+P l


############# figures:

pd_delta_carb = ggplot() +
  geom_col(aes(x = Carbamazepine,
               y = avg_pd_C),
           fill = "lightblue4",
           data = pd_delta) +
  geom_errorbar(data = pd_delta, aes(x = Carbamazepine,
                                     y = avg_pd_C,
                                     ymin = avg_pd_C-sd_pd_C,
                                     ymax = avg_pd_C+sd_pd_C)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-400, 300)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_delta_carb, filename = "Figures/percent_difference_control_delta/pd_delta_carb.png",
       device = "png", height = 7, width = 11)

pd_delta_dic = ggplot() +
  geom_col(aes(x = Diclofenac,
               y = avg_pd_D),
           fill = "lightblue4",
           data = pd_delta) +
  geom_errorbar(data = pd_delta, aes(x = Diclofenac,
                                     y = avg_pd_D,
                                     ymin = avg_pd_D-sd_pd_D,
                                     ymax = avg_pd_D+sd_pd_D)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-400, 300)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_delta_dic, filename = "Figures/percent_difference_control_delta/pd_delta_dic.png",
       device = "png", height = 7, width = 11)

pd_delta_PFOS = ggplot() +
  geom_col(aes(x = PFOS,
               y = avg_pd_P),
           fill = "lightblue4",
           data = pd_delta) +
  geom_errorbar(data = pd_delta, aes(x = PFOS,
                                     y = avg_pd_P,
                                     ymin = avg_pd_P-sd_pd_P,
```

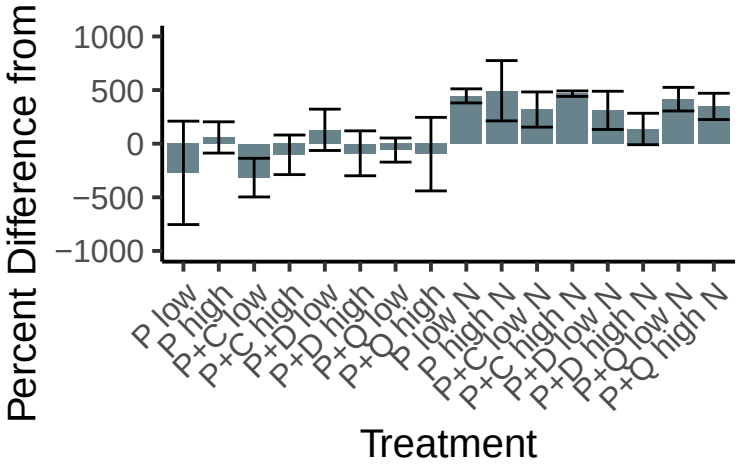
```
                                ymax = avg_pd_P+sd_pd_P)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-1000, 1000)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_delta_PFOS, filename = "Figures/percent_difference_control_delta/pd_delta_PFOS.png",
       device = "png", height = 7, width = 11)

pd_delta_Q = ggplot() +
  geom_col(aes(x = Quinone,
               y = avg_pd_Q),
           fill = "lightblue4",
           data = pd_delta) +
  geom_errorbar(data = pd_delta, aes(x = Quinone,
                                y = avg_pd_Q,
                                ymin = avg_pd_Q-sd_pd_Q,
                                ymax = avg_pd_Q+sd_pd_Q)) +
  xlab("Treatment") +
  ylab("Percent Difference from Control") +
  theme_classic(base_size = 15) +
  scale_y_continuous(limits = c(-3000, 1000)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(pd_delta_Q, filename = "Figures/percent_difference_control_delta/pd_delta_Q.png",
       device = "png", height = 7, width = 11)

pd_delta_carb
```



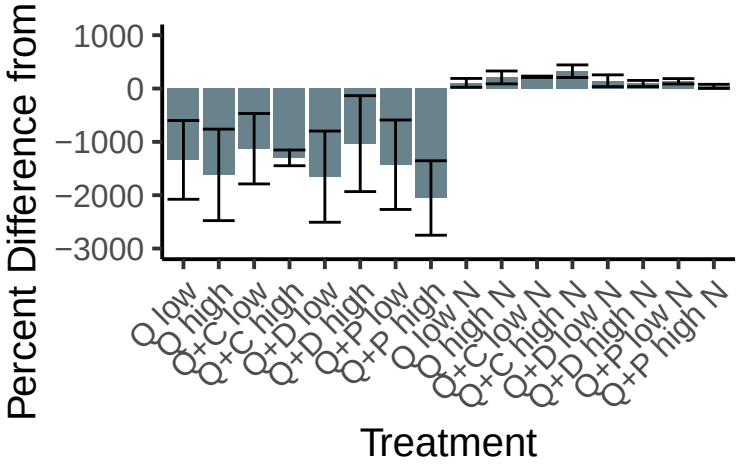

```
pd_delta_dic
```




```
pd_delta_PFOS
```




```
pd_delta_Q
```



```r
############# import data:

Carb_RFU = read_excel("2025_combinations_master.xlsx", sheet = "Carb_RFU")
```

Change in RFU over time

```
## New names:
## * `` -> `...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
## * `` -> `...15`
## * `` -> `...27`
## * `` -> `...28`
## * `` -> `...29`
## * `` -> `...30`
## * `` -> `...37`
## * `` -> `...38`
## * `` -> `...39`
```

```r
Dic_RFU = read_excel("2025_combinations_master.xlsx", sheet = "Dic_RFU")
```

```
## New names:
## * `` -> `...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
## * `` -> `...15`
## * `` -> `...27`
## * `` -> `...28`
## * `` -> `...29`
## * `` -> `...30`
## * `` -> `...37`
## * `` -> `...38`
## * `` -> `...39`
```

```r
PFOS_RFU = read_excel("2025_combinations_master.xlsx", sheet = "PFOS_RFU")
```

```
## New names:
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
## * `` -> `...15`
## * `` -> `...27`
## * `` -> `...28`
## * `` -> `...29`
## * `` -> `...30`
## * `` -> `...37`
## * `` -> `...38`
## * `` -> `...39`
```

```r
Q_RFU = read_excel("2025_combinations_master.xlsx", sheet = "Q_RFU")
```

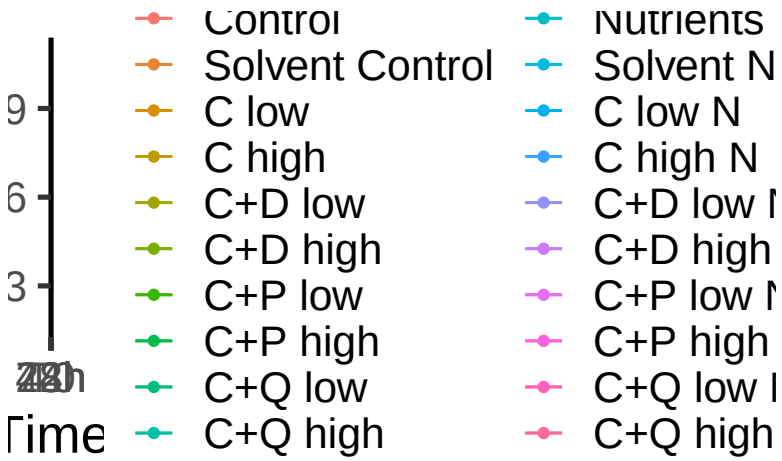
```
## New names:
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
## * `` -> `...15`
## * `` -> `...27`
## * `` -> `...28`
## * `` -> `...29`
## * `` -> `...30`
## * `` -> `...37`
## * `` -> `...38`
## * `` -> `...39`
```

```r
############# mean and st dev:

Carb_change = Carb_RFU %>%
  group_by(Group) %>%
  summarize(mean_T0 = mean(T0, na.rm = TRUE),
            mean_24h = mean(`24h (closer to 27)`, na.rm = TRUE),
            mean_48h = mean(`48h`, na.rm = TRUE),
            mean_72h = mean(`72h`, na.rm = TRUE),
            sd_T0 = sd(T0, na.rm = TRUE),
            sd_24h = sd(`24h (closer to 27)`, na.rm = TRUE),
            sd_48h = sd(`48h`, na.rm = TRUE),
            sd_72h = sd(`72h`, na.rm = TRUE))

Dic_change = Dic_RFU %>%
  group_by(Group) %>%
  summarize(mean_T0 = mean(T0, na.rm = TRUE),
            mean_24h = mean(`24h`, na.rm = TRUE),
            mean_48h = mean(`48h`, na.rm = TRUE),
            mean_72h = mean(`72h`, na.rm = TRUE),
            sd_T0 = sd(T0, na.rm = TRUE),
            sd_24h = sd(`24h`, na.rm = TRUE),
            sd_48h = sd(`48h`, na.rm = TRUE),
            sd_72h = sd(`72h`, na.rm = TRUE))

PFOS_change = PFOS_RFU %>%
  group_by(Group) %>%
  summarize(mean_T0 = mean(T0, na.rm = TRUE),
            mean_24h = mean(`24h`, na.rm = TRUE),
            mean_48h = mean(`48h`, na.rm = TRUE),
            mean_72h = mean(`72h`, na.rm = TRUE),
            sd_T0 = sd(T0, na.rm = TRUE),
            sd_24h = sd(`24h`, na.rm = TRUE),
            sd_48h = sd(`48h`, na.rm = TRUE),
            sd_72h = sd(`72h`, na.rm = TRUE))

Q_change = Q_RFU %>%
  group_by(Group) %>%
  summarize(mean_T0 = mean(T0, na.rm = TRUE),
```

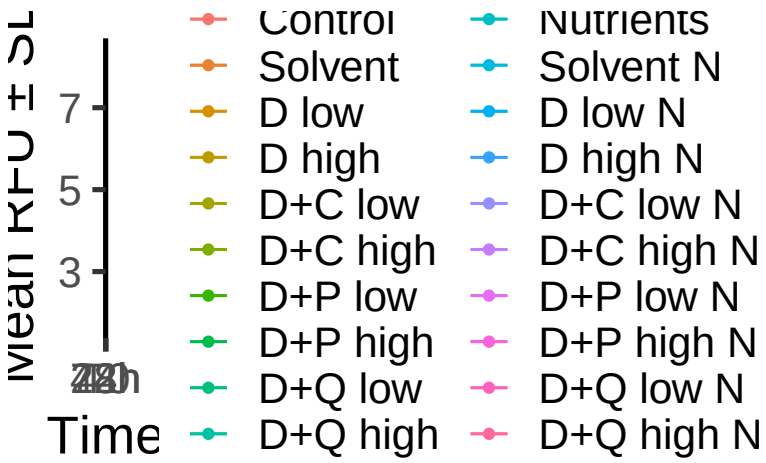
```
            mean_24h = mean(`24h`, na.rm = TRUE),
            mean_48h = mean(`48h`, na.rm = TRUE),
            mean_72h = mean(`72h`, na.rm = TRUE),
            sd_T0 = sd(T0, na.rm = TRUE),
            sd_24h = sd(`24h`, na.rm = TRUE),
            sd_48h = sd(`48h`, na.rm = TRUE),
            sd_72h = sd(`72h`, na.rm = TRUE))

############# rearrange and reformat data:

Dic_change_2 <- Dic_change %>%
  pivot_longer(
    cols = -Group,
    names_to = c(".value", "Time"),
    names_pattern = "(mean|sd)_(.*)"
  )

Dic_change_2$Time <- factor(Dic_change_2$Time, levels = c("T0", "24h", "48h", "72h"))

Dic_change_2$Group <- factor(Dic_change_2$Group,
                             levels = c("control", "solvent control", "Diclow", "Dichigh", "Dic+Carblow
                             labels = c("Control", "Solvent", "D low", "D high", "D+C low", "D+C high",


Carb_change_2 <- Carb_change %>%
  pivot_longer(
    cols = -Group,
    names_to = c(".value", "Time"),
    names_pattern = "(mean|sd)_(.*)"
  )

Carb_change_2$Time <- factor(Carb_change_2$Time, levels = c("T0", "24h", "48h", "72h"))

Carb_change_2$Group <- factor(Carb_change_2$Group,
                              levels = c("control", "solvent control", "Carblow", "Carbhigh", "Carb+Dic
                              labels = c("Control", "Solvent Control", "C low", "C high", "C+D low", "C-



PFOS_change_2 <- PFOS_change %>%
  pivot_longer(
    cols = -Group,
    names_to = c(".value", "Time"),
    names_pattern = "(mean|sd)_(.*)"
  )

PFOS_change_2$Time <- factor(PFOS_change_2$Time, levels = c("T0", "24h", "48h", "72h"))

PFOS_change_2$Group <- factor(PFOS_change_2$Group,
                              levels = c("control", "solvent control", "PFOSlow", "PFOShigh", "PFOS+Car
                              labels = c("Control", "Solvent", "P low", "P high", "P+C low", "P+C high"

Q_change_2 <- Q_change %>%
```

```r
  pivot_longer(
    cols = -Group,
    names_to = c(".value", "Time"),
    names_pattern = "(mean|sd)_(.*)"
  )

Q_change_2$Time <- factor(Q_change_2$Time, levels = c("T0", "24h", "48h", "72h"))

Q_change_2$Group <- factor(Q_change_2$Group,
                              levels = c("control", "solvent control", "6ppdqlow", "6ppdqhigh", "6ppdq+C
                              labels = c("Control", "Solvent", "Q low", "Q high", "Q+C low", "Q+C high"



############# figures:

###Diclofenac

Dic_change_over_time = ggplot(Dic_change_2, aes(x = Time, y = mean, group = Group, color = Group)) +
  geom_line() +
  geom_point() +
  geom_errorbar(aes(ymin = mean - sd, ymax = mean + sd), width = 0.2) +
  labs(
    y = "Mean RFU ± SD",
    x = "Time"
  ) +
  guides(color = guide_legend(ncol = 2)) +
  theme(legend.position = "bottom") +
  theme_classic(base_size = 20)
ggsave(Dic_change_over_time, filename = "Figures/Change_over_time/Dic_change_over_time.png",
       device = "png", height = 7, width = 11)

###Carbamazepine

Carb_change_over_time = ggplot(Carb_change_2, aes(x = Time, y = mean, group = Group, color = Group)) +
  geom_line() +
  geom_point() +
  geom_errorbar(aes(ymin = mean - sd, ymax = mean + sd), width = 0.2) +
  labs(
    y = "Mean RFU ± SD",
    x = "Time"
  ) +
  guides(color = guide_legend(ncol = 2)) +
  theme(legend.position = "bottom") +
  theme_classic(base_size = 20)
ggsave(Carb_change_over_time, filename = "Figures/Change_over_time/Carb_change_over_time.png",
       device = "png", height = 7, width = 11)


###PFOS

PFOS_change_over_time = ggplot(PFOS_change_2, aes(x = Time, y = mean, group = Group, color = Group)) +
  geom_line() +
```

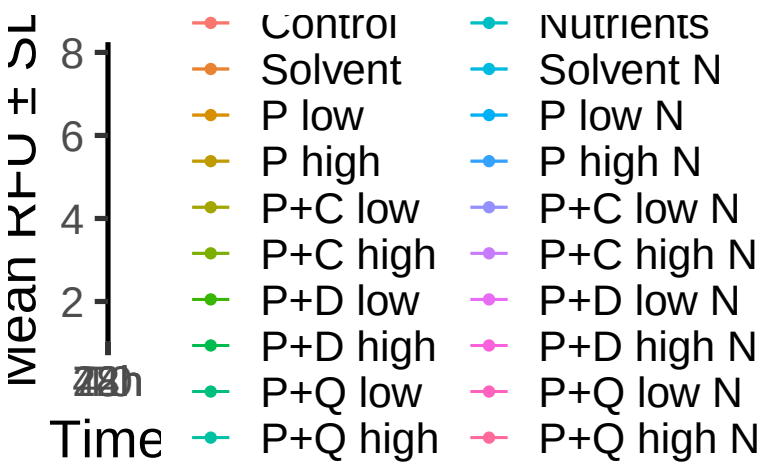
```r
  geom_point() +
  geom_errorbar(aes(ymin = mean - sd, ymax = mean + sd), width = 0.2) +
  labs(
    y = "Mean RFU ± SD",
    x = "Time"
  ) +
  guides(color = guide_legend(ncol = 2)) +
  theme(legend.position = "bottom") +
  theme_classic(base_size = 20)
ggsave(PFOS_change_over_time, filename = "Figures/Change_over_time/PFOS_change_over_time.png",
       device = "png", height = 7, width = 11)


###6ppdq

Q_change_over_time = ggplot(Q_change_2, aes(x = Time, y = mean, group = Group, color = Group)) +
  geom_line() +
  geom_point() +
  geom_errorbar(aes(ymin = mean - sd, ymax = mean + sd), width = 0.2) +
  labs(
    y = "Mean RFU ± SD",
    x = "Time"
  ) +
  guides(color = guide_legend(ncol = 2)) +
  theme(legend.position = "bottom") +
  theme_classic(base_size = 20)
ggsave(Q_change_over_time, filename = "Figures/Change_over_time/Q_change_over_time.png",
       device = "png", height = 7, width = 11)



combined_time = ggarrange(Carb_change_over_time + rremove("ylab") + rremove("xlab"), Dic_change_over_tim
                          labels = c("A", "B", "C", "D"),
                          label.x = 0.04,
                          label.y = 0.99,
                          nrow = 2,
                          ncol = 2)



combined_time  = annotate_figure(combined_time, left = textGrob("Chl a Fluorescence (RFU)", rot = 90, v
                                 bottom = textGrob("Time (h)", gp = gpar(cex = 1.3, fontsize = 18)))


Carb_change_over_time
```



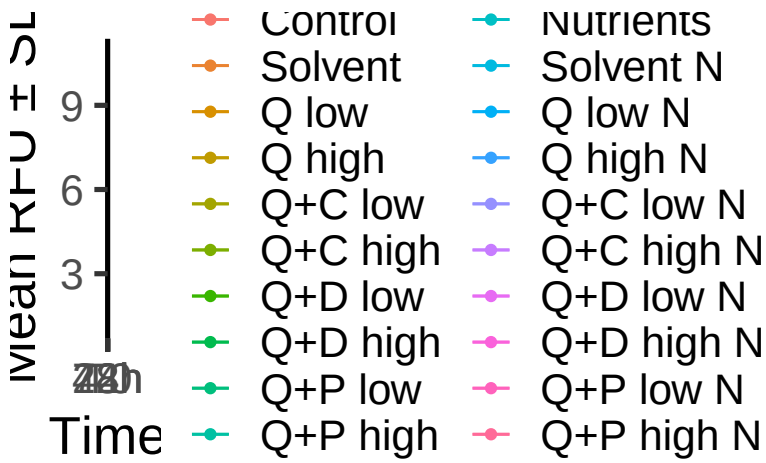

```
Dic_change_over_time
```



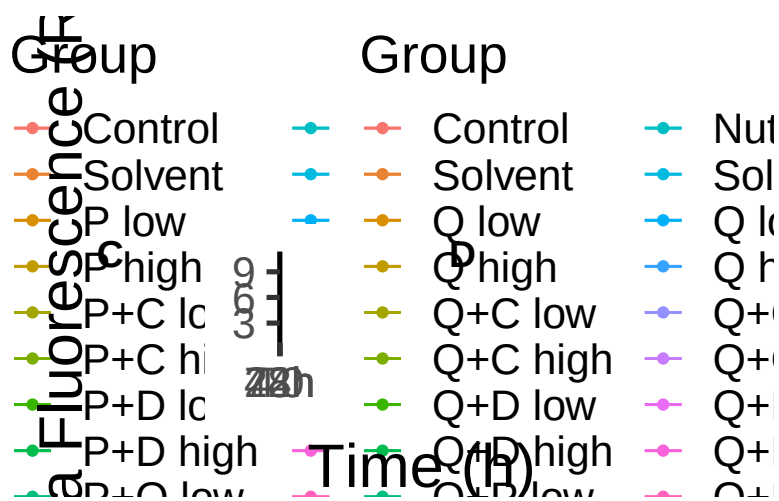

```
PFOS_change_over_time
```



```
Q_change_over_time
```




```
combined_time
```




```
############# import data:

all_pam = read_excel("2025_combinations_master.xlsx", sheet = "all_FvFm")

glimpse(all_pam)
```

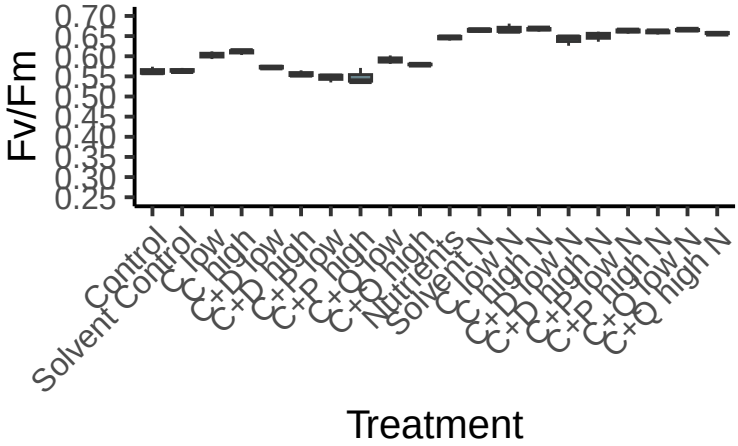
Fv/Fm at 72h

```
## Rows: 60
## Columns: 8
## $ Carbamazepine <chr> "control", "control", "control", "solvent control", "sol~
## $ FvFm_C        <dbl> 0.560, 0.574, 0.555, 0.563, 0.562, 0.571, 0.647, 0.639, ~
## $ Diclofenac    <chr> "control", "control", "control", "solvent control", "sol~
```

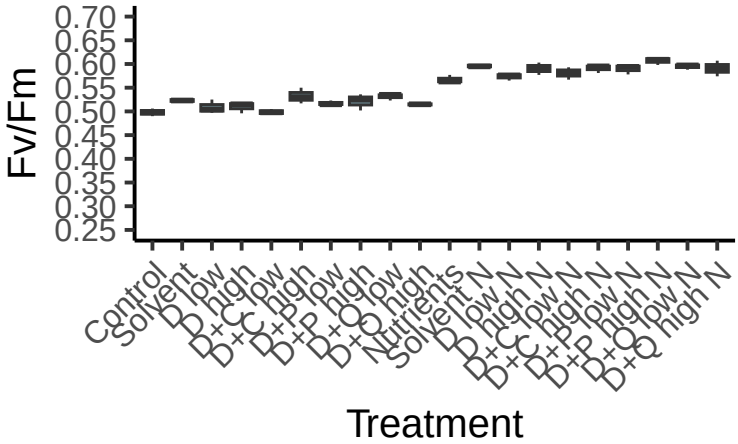
```
## $ FvFm_D        <dbl> 0.490, 0.506, 0.498, 0.523, 0.527, 0.519, 0.563, 0.577, ~
## $ PFOS          <chr> "control", "control", "control", "solvent control", "sol~
## $ FvFm_P        <dbl> 0.405, 0.428, 0.434, 0.312, 0.323, 0.316, 0.508, 0.506, ~
## $ Quinone       <chr> "control", "control", "control", "solvent control", "sol~
## $ FvFm_Q        <dbl> 0.525, 0.537, 0.535, 0.547, 0.553, 0.569, 0.608, 0.597, ~
```

```
############# rearrange and reformat data:


all_pam$Carbamazepine <- factor(all_pam$Carbamazepine,
                                levels = c("control", "solvent control", "Carblow", "Carbhigh", "Carb+D
                                labels = c("Control", "Solvent Control", "C low", "C high", "C+D low",

all_pam$Diclofenac <- factor(all_pam$Diclofenac,
                             levels = c("control", "solvent control", "Diclow", "Dichigh", "Dic+Carblow
                             labels = c("Control", "Solvent", "D low", "D high", "D+C low", "D+C high",


all_pam$PFOS <- factor(all_pam$PFOS,
                       levels = c("control", "solvent control", "PFOSlow", "PFOShigh", "PFOS+Carblow",
                       labels = c("Control", "Solvent", "P low", "P high", "P+C low", "P+C high", "P+D

all_pam$Quinone <- factor(all_pam$Quinone,
                          levels = c("control", "solvent control", "6ppdqlow", "6ppdqhigh", "6ppdq+Carl
                          labels = c("Control", "Solvent", "Q low", "Q high", "Q+C low", "Q+C high", "(

############# figures:

carb_pam = ggplot() +
  geom_boxplot(aes(x = Carbamazepine,
                   y = FvFm_C),
               fill = "lightblue4",
               data = all_pam) +
  xlab("Treatment") +
  ylab("Fv/Fm") +
  scale_y_continuous(breaks = seq(0.25, 0.7, by = 0.05), limits = c(0.25,0.7)) +
  theme_classic(base_size = 15) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(carb_pam, filename = "Figures/FvFm/carb_pam.png",
       device = "png", height = 7, width = 11)


dic_pam = ggplot() +
  geom_boxplot(aes(x = Diclofenac,
                   y = FvFm_D),
               fill = "lightblue4",
               data = all_pam) +
  xlab("Treatment") +
  ylab("Fv/Fm") +
  scale_y_continuous(breaks = seq(0.25, 0.7, by = 0.05), limits = c(0.25,0.7)) +
  theme_classic(base_size = 15) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(dic_pam, filename = "Figures/FvFm/dic_pam.png",
       device = "png", height = 7, width = 11)
```

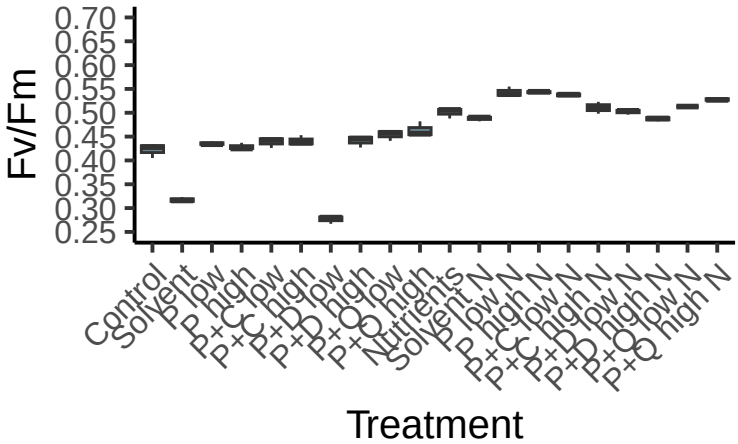
```
PFOS_pam = ggplot() +
  geom_boxplot(aes(x = PFOS,
                   y = FvFm_P),
               fill = "lightblue4",
               data = all_pam) +
  xlab("Treatment") +
  ylab("Fv/Fm") +
  scale_y_continuous(breaks = seq(0.25, 0.7, by = 0.05), limits = c(0.25,0.7)) +
  theme_classic(base_size = 15) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(PFOS_pam, filename = "Figures/FvFm/PFOS_pam.png",
       device = "png", height = 7, width = 11)

Q_pam = ggplot() +
  geom_boxplot(aes(x = Quinone,
                   y = FvFm_Q),
               fill = "lightblue4",
               data = all_pam) +
  xlab("Treatment") +
  ylab("Fv/Fm") +
  scale_y_continuous(breaks = seq(0.25, 0.7, by = 0.05), limits = c(0.25,0.7)) +
  theme_classic(base_size = 15) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(Q_pam, filename = "Figures/FvFm/Q_pam.png",
       device = "png", height = 7, width = 11)

carb_pam
```

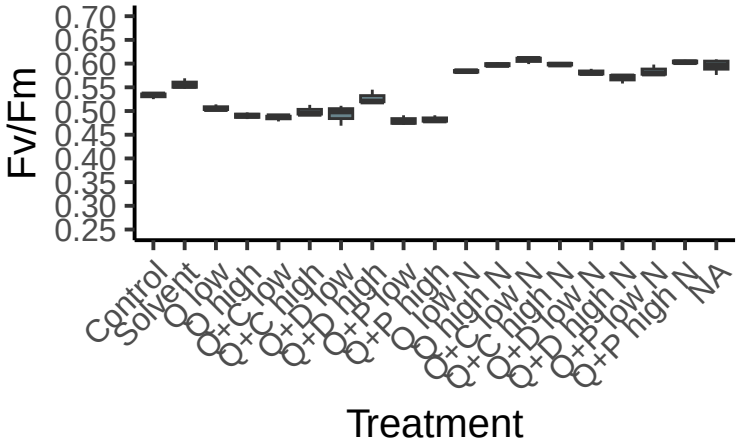



```
dic_pam
```




```
PFOS_pam
```

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Q_pam
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